MIL-T-46573B(MU)
24 January 1968
SUPERSEDING
MIL-T-46573A (ORD)
27 June 1962

MILITARY SPECIFICATION
THRUSTER, CARTRIDGE ACTUATED, M5A2
METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Thruster, Cartridge Actuated, M5A2, Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunitions and Special Weapons, General Specification for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber O-Rings
MIL-STD-639	- Surface Roughness Values Preferred for Army Ordnance Materiel
MIL-STD-1167	- Ammunition Data Card

DRAWINGS

Munitions Command

F8594747	- Thruster, Cartridge Actuated, M5A2, Metal Parts Assembly

FSC 1377

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2.1 DRAWINGS (Cont'd)

Munitions Command

C8797741	- Container, Ammunition, Fiber, M347, for Thruster, Cartridge Actuated, M20A1, XM25 or M5A2
C8797742	- Box, Packing, Ammunition, for Thruster, Cartridge Actuated, M20A1, XM25 or M5A2 in Fiber Container, M347
IEL-8594747	- Inspection Equipment List for Thruster, Cartridge Actuated, M5A2, Metal Parts Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Material.- The contractor shall be responsible for selecting from the classification list contained in the raw material specification, the surface and physical condition of the raw material intended for use in manufacture.

3.3 Physical and mechanical properties.- Physical and mechanical properties required by the drawing shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.4 Hydrostatic pressure.- The body and head shall withstand a constant internal pressure of 8000 psi, minimum, for 15 seconds, minimum, without application of any external load and shall show no evidence of leakage or mechanical failure (see 6.1).

3.5 Structural strength.- The assembled thruster shall withstand a static tensile force of 2000 pounds, minimum, for 15 seconds, minimum, between mounting points without unlocking and without mechanical failure (see 6.1).

3.6 After application of protective finish, the shear pin shall shear when a load of 48 $\pm$ 4 lbs. is applied at the steady rate of 0.125 inches per minute.

3.7 Metal parts assembly.- Parts shall be combined to produce an assembly which conforms to the requirements of the Metal Parts Assembly Drawing F8594747.

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3.7.1 The head and body shall be assembled handtight.

3.7.2 Trunnion rotation.- Upon completion of assembly, the trunnion shall be free to rotate 360 degrees.

3.7.3 End sleeve rotation.- Upon completion of assembly and after structural strength test, the end sleeve shall be free to rotate 360 degrees in a clockwise direction.

3.8 Protective finish.- Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of contour configuration.

3.9 Residual magnetism.- The metal parts assembly shall not deflect the indicator of the compass more than three degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.10 Workmanship.- The requirements for workmanship are as specified in the applicable drawings, referenced specifications and the following:

3.10.1 Metal defects.- All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.10.2 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.10.3 Foreign matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspection are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference is made to MIL-STD-109 to define quality assurance terms used.

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4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550 except as specified herein.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 5 metal parts assemblies, 10 sets of parts and fifty shear pins from the same lot of shear pins tested by the contractor. The samples shall be subjected to tests in 4.3.3. The test procedures shall be in accordance with 4.4. The samples shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, must be the same as used in regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- Unless otherwise specified by the contracting officer, formation of lots and presentation of lots shall be in accordance with the "Submission of Product" provisions of MIL-STD-105.

4.3.1.2 Size of lot.- The shipping lot of metal parts assemblies shall consist of 1200 thrusters, maximum, provided the homogeneity requirements of inspection lots have been retained.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans and associated criteria in Table I and II. For an individual defect (single characteristic), the criteria in Table I shall be used. For total defects within a major or minor group, the criteria in Table II shall be used. All nonconforming material shall be rejected. The Ordnance Standard Finish Blocks prescribed in MIL-STD-639 shall

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4.3.2 (Cont'd)

be used as a basis of comparison for surface roughness determination. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE I

Lot Size		Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1,300	110	1	2	2	3
1,301 - 3,200	150	2	3	3	4
3,201 - 8,000	225	3	4	4	5
8,001 - 22,000	300	4	5	5	6

TABLE II

Total Allowable Defects per Sample Size

Sample Sizes for Major Defects	110	150	225	300
Number of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14
Sample Sizes for Minor Defects	**110**	**150**	**225**	**300**
Number of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

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4.3.2.1 Pin, Shear, Drawing B8593284 covering a detail of Drawing F8594747

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Diameter, max.	Gage
102.	Total length, min.	Gage
103.	Metal defective	Visual
104.	Protective finish inadequate or defective	Visual
105.	Surface finish, improper	Visual
106.	Burr	Visual
107.	Foreign matter	Visual
MINOR		
201.	Total length, max.	Gage
202.	Profile of tapered end	Gage

4.3.2.2 Key, Drawing C8593857 covering a detail of Drawing F8594747

CRITICAL		
	None defined	
MAJOR		
101.	Radial profile of length	Gage
102.	Radial profile of width	Gage
103.	Metal defective	Visual
104.	Protective finish inadequate or defective	Visual
105.	Surface finish improper	Visual
106.	Burr or sharp corners	Visual
107.	Foreign matter	Visual
MINOR		
201.	Total length	Gage
202.	Small radii incorrect	Gage

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4.3.2.3 Guide, Firing Pin, Drawing D8593863 covering a detail of Drawing F8594747

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Concentricity of striker hole with cavity	Gage
102.	Concentricity of body with cavity	Gage
103.	Concentricity of boss cavity with cavity	Gage
104.	True position of pin hole in body with cavity	Gage
105.	Diameter of cavity, max.	Gage
106.	Diameter of body, min.	Gage
107.	Diameter of datum location	Gage
108.	Diameter of shear pin hole, max.	Gage
109.	Height of boss	Gage
110.	Angle of boss	Gage
111.	Radius at root of boss	Gage
112.	Escapement hole missing	Visual
113.	Metal defective	Visual
114.	Protective finish inadequate or defective	Visual
115.	Surface finish of cavity and face, improper	Visual
116.	Burr or sharp corners	Visual
117.	Foreign matter	Visual
MINOR		
201.	Diameter of cavity, min.	Gage
202.	Diameter of body, max.	Gage
203.	Diameter of shear pin hole, min.	Gage
204.	Diameter of striker hole	Gage
205.	Diameter of boss cavity	Gage
206.	Length of striker hole	Gage
207.	Length from bottom of cavity to face of boss	Gage
208.	Total length	Gage
209.	Location of shear pin hole	Gage
210.	Diameter of escapement hole	Gage
211.	Location of escapement hole	Gage
212.	Marking missing, incorrect or illegible	Visual
213.	Radius, chamfer or both missing or incorrect, except as otherwise classified	Visual
214.	Surface finish improper except as otherwise classified	Visual

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4.3.2.4 Pin, Firing, Drawing D8593864 covering a detail of Drawing F8594747

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Concentricity of groove, with body	Gage
102.	Concentricity of striker with body	Gage
103.	Concentricity of striker shoulder with body	Gage
104.	True position of small hole with body	Gage
105.	Diameter of groove	Gage
106.	Diameter of body, min.	Gage
107.	Diameter of striker, max.	Gage
108.	Diameter of pin hole, max.	Gage
109.	Depth of pin hole, max.	Gage
110.	Profile of striker	Gage
111.	Profile of groove	Gage
112.	Metal defective	Visual
113.	Protective finish inadequate or defective	Visual
114.	Surface finish of body and striker tip, improper	Visual
115.	Burr or sharp corner	Visual
116.	Foreign matter	Visual
MINOR		
201.	Diameter of body, max.	Gage
202.	Diameter of striker, min.	Gage
203.	Diameter of pin hole, min.	Gage
204.	Diameter of striker shoulder	Gage
205.	Depth of pin hole, min.	Gage
206.	Length from body to tip of striker	Gage
207.	Length of striker	Gage
208.	Total length	Gage
209.	Location of pin hole	Gage
210.	Location of groove	Gage
211.	Depth of slot in base	Gage
212.	Length of body diameter forward of groove, min.	Gage
213.	Width of groove	Gage
214.	Width of slot in base	Gage

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4.3.2.4 (Cont'd)

Defect	Method of Inspection
MINOR (Cont'd)	
215. Location of slot in base	Gage
216. Radius incorrect	Gage
217. Marking missing, incorrect or illegible	Visual
218. Surface finish improper except as otherwise classified	Visual

4.3.2.5 Body, Drawing D8594480 covering a detail of Drawing F8594747

CRITICAL

None defined

MAJOR

101. Pitch diameter of thread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Concentricity of pitch diameter of thread with large bore	Gage
104. Concentricity of small bore with large bore	Gage
105. Concentricity of groove with small bore	Gage
106. Concentricity of counterbore with small bore	Gage
107. Diameter of large bore, max.	Gage
108. Diameter of counterbore	Gage
109. Diameter of key groove	Gage
110. Diameter of small bore	Gage
111. Diameter of body, thread end	Gage
112. Diameter of internal undercut, max. (if applicable)	Gage
113. Location of key groove	Gage
114. Straightness of bore	Gage
115. Diameter of thread undercut, min.	Gage
116. Profile of key groove	Gage
117. Thread damaged or not full	Visual
118. Metal defective	Visual
119. Protective finish inadequate or defective	Visual
120. Surface finish of large bore, improper	Visual
121. Burr or sharp corner	Visual
122. Foreign matter	Visual

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4.3.2.5 (Cont'd)

	Defect	Method of Inspection
MINOR		
201.	Pitch diameter of thread, max.	Gage
202.	Diameter of large bore, min.	Gage
203.	Diameter of body, max.	Gage
204.	Depth of counterbore	Gage
205.	Total length	Gage
206.	Length to exterior shoulder	Gage
207.	Width across flats	Gage
208.	Thickness of wall (for length indicated)	Gage
209.	Diameter of shoulder	Gage
210.	Length to base of large bore	Gage
211.	Length of shoulder	Gage
212.	Width of thread undercut, max.	Gage
213.	Width of internal undercut, max. (if applicable)	Gage
214.	Location of thread undercut	Gage
215.	Width of flat	Gage
216.	Location of flats, incorrect	Gage
217.	Radius or chamfers incorrect	Gage
218.	Marking missing, incorrect or illegible	Visual
219.	Surface finish improper, except as otherwise classified	Visual

4.3.2.6 Trunnion, Drawing D8594482 covering a detail of Drawing F8594747

CRITICAL

None defined

MAJOR

101.	Perpendicularity of sides with inside diameter	Gage
102.	Concentricity of lugs with each other	Gage
103.	True position of lugs with inside diameter	Gage
104.	Symmetry of flats at root of lugs with inside diameter	Gage
105.	Diameter of lug, min.	Gage
106.	Width across flats, min.	Gage
107.	Thickness of wall, min.	Gage
108.	Radius at base of lugs incorrect	Gage
109.	Metal defective	Visual
110.	Protective finish inadequate or defective	Visual

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4.3.2.6 (Cont'd)

Defect		Method of Inspection
MINOR		
201.	Diameter of lug, max.	Gage
202.	Diameter, outside, max.	Gage
203.	Diameter inside	Gage
204.	Width	Gage
205.	Length of lug	Gage
206.	Width across flats, max.	Gage
207.	Thickness of wall, max.	Gage
208.	Location of lug	Gage
209.	Radius or chamfers incorrect	Gage
210.	Marking missing, incorrect or illegible	Visual
211.	Surface finish improper	Visual
212.	Burr or sharp corner	Visual
213.	Foreign matter	Visual

4.3.2.7 Retainer, Cartridge, Drawing D8594483 covering a detail of Drawing F8594747

CRITICAL

None defined

MAJOR

101.	Pitch diameter of thread, min.	Gage
102.	Major diameter of thread, min.	Gage
103.	Concentricity of large cavity with small cavity	Gage
104.	Concentricity of groove with flange	Gage
105.	Concentricity of flange with pitch diameter	Gage
106.	Concentricity of small cavity with pitch diameter	Gage
107.	Diameter of flange, min.	Gage
108.	Diameter of groove	Gage
109.	Diameter of small cavity, max.	Gage
110.	Diameter of large cavity, max.	Gage
111.	Length of effective thread, min.	Gage
112.	Internal radius	Gage
113.	Profile of groove	Visual
114.	Number of elongated slots incorrect	Visual
115.	Escapement hole at rear of flange missing	Visual
116.	Thread damaged or not full	Visual
117.	Metal defective	Visual
118.	Protective finish inadequate or defective	Visual
119.	Burr or sharp corner	Visual
120.	Foreign matter	Visual

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4.3.2.7 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Pitch diameter of thread, max.	Gage
202. Diameter of flange, max.	Gage
203. Diameter of large cavity, min.	Gage
204. Diameter of small cavity, min.	Gage
205. Diameter of rear of thread	Gage
206. Diameter of small end	Gage
207. Depth of large cavity	Gage
208. Depth to bottom of small cavity	Gage
209. Width of elongated slot	Gage
210. Diameter of escapement hole	Gage
211. Depth of base slot	Gage
212. Total length	Gage
213. Length of flange	Gage
214. Length of elongated slot	Gage
215. Length to end of flange	Gage
216. Width of groove	Gage
217. Width of base slot	Gage
218. Location of groove	Gage
219. Location of elongated slot	Gage
220. Location of escapement hole	Gage
221. Location of base slot	Gage
222. Radius or chamfers incorrect	Gage
223. Marking missing, incorrect or illegible	Visual
224. Surface finish improper	Visual

4.3.2.8 Buffer, Drawing D8594485 covering a detail of Drawing F8594747

CRITICAL

None defined

MAJOR

101. Pitch diameter of thread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Concentricity of flange with thread	Gage
104. Diameter of groove	Gage
105. Diameter of flange, groove end, min.	Gage
106. Diameter of head, min.	Gage
107. Diameter of body, min.	Gage

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4.3.2.8 (Cont'd)

Defect	Method of Inspection
MAJOR (Cont'd)	
108. Diameter of flange, thread end, min.	Gage
109. Diameter of thread undercut, min.	Gage
110. Length of thread end, min.	Gage
111. Length of body from thread flange to head, min.	Gage
112. Width of groove	Gage
113. Profile of groove	Gage
114. Width of thread undercut	Gage
115. Thread damaged or not full	Visual
116. Metal defective	Visual
117. Protective finish inadequate or defective	Visual
118. Burr or sharp corner	Visual
119. Foreign matter	Visual
MINOR	
201. Pitch diameter of thread, max.	Gage
202. Diameter of head, max.	Gage
203. Diameter of body, max.	Gage
204. Diameter of flange, groove end, max.	Gage
205. Diameter of flange, thread end, max.	Gage
206. Length of threaded end, max.	Gage
207. Length of body from thread flange to head, max.	Gage
208. Total length	Gage
209. Radius or chamfers incorrect	Gage
210. Thickness of flange, groove end	Gage
211. Width of flange, thread end	Gage
212. Slot width	Gage
213. Slot depth	Gage
214. Slot location	Gage
215. Surface finish, improper	Visual
216. Marking missing, incorrect or illegible	Visual

4.3.2.9 Sleeve, End, Drawing D8594488 covering a detail of Drawing F8594747

CRITICAL

None defined

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4.3.2.9 (Cont'd)

Defect		Method of Inspection
MAJOR		
101.	Pitch diameter of thread, min.	Gage
102.	Major diameter of thread, min.	Gage
103.	Concentricity of shoulder with body	Gage
104.	Concentricity of large cavity with body	Gage
105.	Concentricity of cavity at midsection with large cavity	Gage
106.	Concentricity of cavity at thread end with large cavity	Gage
107.	Symmetry of slots with body	Gage
108.	Diameter of body, min.	Gage
109.	Diameter of large cavity, max.	Gage
110.	Diameter of cavity, thread end, max.	Gage
111.	Diameter of hole through midsection	Gage
112.	Diameter of shoulder, min.	Gage
113.	Width of slots	Gage
114.	Location of slots	Gage
115.	Spherical radius at thread end	Gage
116.	Thread damaged or not full	Visual
117.	Metal defective	Visual
118.	Protective finish inadequate or defective	Visual
119.	Burr or sharp corner	Visual
120.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of thread, max.	Gage
202.	Diameter of body, max.	Gage
203.	Diameter of large cavity, min.	Gage
204.	Diameter of cavity, thread end, min.	Gage
205.	Diameter of shoulder, max.	Gage
206.	Diameter of flange	Gage
207.	Diameter of thread undercut, min.	Gage
208.	Length of slot	Gage
209.	Length of small hole through midsection	Gage
210.	Variation in alignment, flange face with face of large cavity seat	Gage
211.	Total length	Gage
212.	Width of shoulder	Gage
213.	Width of thread undercut	Gage
214.	Width across flats	Gage
215.	Width of flange at flat	Gage

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4.3.2.9 (Cont'd)

Defect	Method of Inspection
MINOR (Cont'd)	
216. Location of undercut	Gage
217. Location of flat	Gage
218. Location of flange	Gage
219. Radius or chamfers incorrect, except as otherwise classified	Gage
220. Marking missing, incorrect or illegible	Visual
221. Surface finish improper	Visual

4.3.2.10 Spring, Piston Locking, Drawing C8594508 covering a detail of Drawing F8594747

CRITICAL

None defined

MAJOR	
101. Load at short compressed length	Gage
102. Load at long compressed length	Gage
103. Length, solid, max.	Gage
104. Ends incorrect	Visual
105. Protective finish inadequate or defective	Visual
106. Burr or sharp edges	Visual
107. Foreign matter	Visual
MINOR	
201. Free fit into hole	Gage
202. Free fit over rod	Gage

4.3.2.11 Head, Drawing D8594749 covering a detail of Drawing F8594747

CRITICAL

None defined

MAJOR	
101. Pitch diameter of small thread, max.	Gage
102. Minor diameter of small thread, max.	Gage

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4.3.2.11 (Cont'd)

	Defect	Method of Inspection
MAJOR (Cont'd)		
103.	Pitch diameter of midsection thread	Gage
104.	Minor diameter of midsection thread	Gage
105.	Pitch diameter of large thread, max.	Gage
106.	Minor diameter of large thread, max.	Gage
107.	Concentricity of pitch diameter of large thread with pitch diameter of midsection thread	Gage
108.	Depth of counterbore, small end	Gage
109.	Depth of cavity, small end	Gage
110.	Length of effective thread, small end, min.	Gage
111.	Length of thread, small end, max.	Gage
112.	Diameter of undercut, large thread, max.	Gage
113.	Diameter of undercut, midsection thread, max.	Gage
114.	Diameter of chamfer at small end	Gage
115.	Width of undercut, large thread	Gage
116.	Width of undercut, midsection thread	Gage
117.	Angles of chamfers at small end	Gage
118.	Thread damaged or not full	Visual
119.	Metal defective	Visual
120.	Protective finish inadequate or defective	Visual
121.	Surface finish of chamfers, small end, improper	Visual
122.	Burr	Visual
123.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of small thread, min.	Gage
202.	Pitch diameter of midsection thread, min.	Gage
203.	Pitch diameter of large thread, min.	Gage
204.	Pitch diameter of set screw thread	Gage
205.	Minor diameter of set screw thread, max.	Gage
206.	Diameter of counterbore at small end	Gage
207.	Diameter of hole between midsection and small end	Gage
208.	Diameter of body midsection	Gage
209.	Depth of large cavity	Gage
210.	Width across flats	Gage
211.	Diameter of body, large end	Gage
212.	Diameter of body, small end	Gage

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4.3.2.11 (Cont'd)

Defect	Method of Inspection
MINOR (Cont'd)	
213. Depth of bottom of midsection	Gage
214. Length of large body diameter	Gage
215. Length of small body diameter	Gage
216. Length of flat	Gage
217. Total length	Gage
218. Location of flat	Gage
219. Radius or chamfers incorrect	Gage
220. Marking missing, incorrect or illegible	Visual
221. Surface finish improper, except as otherwise classified	Visual

4.3.2.12 Piston, Drawing F8594750 covering a detail of Drawing F8594747

CRITICAL

None defined

MAJOR

101. Pitch diameter of thread in small cavity, max.	Gage
102. Minor diameter of thread in small cavity, max.	Gage
103. Concentricity of midsection with body, large end	Gage
104. Concentricity of small diameter of body with body, large end	Gage
105. Concentricity of ring groove with body, large end	Gage
106. Concentricity of pitch diameter of thread with body, small end	Gage
107. Concentricity of cavity with body, small end	Gage
108. Concentricity of key escape groove with body, small end	Gage
109. Concentricity of key locking groove with body, small end	Gage
110. Diameter of body, large end, min.	Gage
111. Diameter of ring groove	Gage
112. Diameter of shoulder, large end, min.	Gage
113. Diameter of body, midsection	Gage
114. Diameter of body, small end	Gage
115. Diameter of key escape groove, max.	Gage

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4.3.2.12 (Cont'd)

	Defect	Method of Inspection
MAJOR (Cont'd)		
116.	Diameter of key locking groove, min.	Gage
117.	Diameter of small cavity	Gage
118.	Depth of large cavity and drill point, max.	Gage
119.	Depth of thread cavity, max.	Gage
120.	Depth of small cavity, max.	Gage
121.	Length of effective thread, min.	Gage
122.	Length to key locking groove, min.	Gage
123.	Length to large body diameter, min.	Gage
124.	Length of shoulder, large end	Gage
125.	Location of angle in key groove	Gage
126.	Length to end of key escape groove	Gage
127.	Length of small body diameter, min.	Gage
128.	Profile of key escape and locking grooves	Gage
129.	Profile of ring groove	Gage
130.	Thread damaged or not full	Visual
131.	Metal defective	Visual
132.	Protective finish inadequate or defective	Visual
133.	Burr or sharp corner	Visual
134.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of thread in small cavity, min.	Gage
202.	Pitch diameter of set screw thread, min.	Gage
203.	Diameter of body, large end, max.	Gage
204.	Diameter of shoulder, large end, max.	Gage
205.	Diameter of key escape groove, min.	Gage
206.	Diameter of key locking groove, max.	Gage
207.	Diameter of undercut, min.	Gage
208.	Depth of small cavity, min.	Gage
209.	Length to body, large end, max.	Gage
210.	Length to key locking groove, max.	Gage
211.	Diameter of undercut, min.	Gage
212.	Diameter of large cavity	Gage
213.	Length to body, midsection, max.	Gage
214.	Total length	Gage
215.	Width of ring groove	Gage
216.	Width of undercut, max.	Gage
217.	Location of ring groove	Gage
218.	Location of set screw hole	Gage
219.	Radii or chamfers incorrect	Gage
220.	Marking missing, incorrect or illegible	Visual
221.	Surface finish improper	Visual

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4.3.2.13 Assembly (Firing Pin, O-Ring and Guide), Drawing F8594747, all component parts except C8593863, C8593864 and MS 29513-12.

	Defect	Method of Inspection
CRITICAL		
1.	Protrusion of firing pin	Gage
MAJOR		
101.	Any part missing	Visual
102.	Thread damaged or not full	Visual
103.	Metal defective	Visual
104.	Protective finish inadequate or defective	Visual
105.	Foreign matter	Visual
MINOR		
	None defined	

4.3.2.14 Assembly, Drawing F8594747

CRITICAL		
	None defined	
MAJOR		
101.	Trunnion not free to rotate 360 degrees	Manual
102.	End sleeve not free to rotate 360 degrees clockwise	Manual
103.	Any part missing except as otherwise classified	Visual
104.	Improper assembly	Visual
105.	Thread damaged or not full	Visual
106.	Marking missing, incorrect or illegible	Visual
107.	Burr	Visual
108.	Foreign matter	Visual
MINOR		
201.	Shipping cap or plug missing	Visual
202.	Protective finish inadequate or defective	Visual

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4.3.2.15 Container, Fiber, Drawing C8797741

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Gap between body and cover, max.	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damaged through all impregnated layers	Visual

4.3.2.16 Box, Packing, Drawing C8797742

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Manual/Visual
203. Strapping missing, loose, broken or improperly assembled	Manual/Visual
204. Contents exposed	Visual
205. Box damaged	Visual
206. Marking misleading or unidentifiable	Visual
207. Number of units incorrect	Visual

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Hardness.- Fifty units from each heat treated lot of components shall be tested for compliance with the values noted on the applicable drawing. Failure of any unit to comply with drawing requirements shall be cause for rejection of the lot of components.

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4.3.3.2 Hydrostatic pressure.- Each body and each head shall be subjected to an internal hydrostatic pressure. Failure of any unit to comply with the requirements of 3.4 shall be cause for rejection of the unit.

4.3.3.3 Structural strength.- Each thruster shall be subjected to the structural strength test. Failure of any thruster to comply with the requirements of 3.5 shall be cause for rejection of the thruster.

4.3.3.4 Shear pin strength.- Fifty pins from each lot shall be tested for shear strength. Failure of any shear pin to comply with the requirements of 3.6 shall be cause for rejection of the lot.

4.3.3.5 Protective finish.- The number of units for protective finish tests shall be in accordance with the applicable specifications. Failure of any sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.3.3.6 Residual magnetism.- Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any unit to comply with the requirements of 3.9 shall be cause for rejection of the assembly.

4.3.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-8594747.

4.4 Test methods and procedures.

4.4.1 Hardness.- Prior to the application of protective finish, a hardness testing machine shall be used to determine the hardness values specified on the detail drawings.

4.4.2 Hydrostatic pressure.- Prior to application of protective finish, the body and head, individually, shall be subjected to an internal hydrostatic pressure. The testing device used shall internally seal the body and head, as applicable, at each end without applying any axial load other than that applied by the liquid pressure.

4.4.3 Structural strength.- The thruster shall be subjected to the structural strength test in accordance with the requirements of 3.5. The test shall be performed in a fixture in conjunction with a tensile testing machine.

4.4.4 Shear pin strength.- After application of protective finish, the load shall be applied at the steady rate of 0.125 inches per minute. The test shall be performed in a fixture in conjunction with a tensile testing machine.

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4.4.5 Protective finish.- Test procedures for protective finish shall be in accordance with the specification referenced on the component drawing.

4.4.6 Residual magnetism.- The thruster shall be tested for excessive residual magnetism in an area fr[illegible] local magnetic effects by placing the thruster 6 inches away from, and at the same height as the compass, in a north-south horizontal position (head end at south) with the compass set in an east-west heading. The thruster shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the thruster.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking - Level A.- Packing and marking shall be in accordance with Drawings C8797741 and C8797742, except that the marking shall reference the M5A2 Thruster.

5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

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Preparing activity:

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 84 WHICH MAY BE USED.

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MILITARY SPECIFICATION

THRUSTER, CARTRIDGE ACTUATED, M5A2
METAL PARTS ASSEMBLY

This amendment forms a part of Military Specification MIL-T-46573B(MU), dated 24 January 1968, and is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1 Delete "MIL-STD-639 Surface Roughness Values Preferred for Army Ordnance Material"

PAGE 2

Add: "2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the document which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (5241), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of the specification or by letter.

AMSC N/A FSC 1377

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ANSI B46.1-78 Surface Texture (Surface Roughness, Waviness and Lay)

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, NY 10018.)

(Nongovernment standards and other publications are normally available from the organization which prepares or which distributes the documents. These documents also may be available in or through libraries or other informational services.)"

PAGE 4

4.3.2, last line, delete: "The Ordnance Standard Finish Blocks prescribed in MIL-STD-639 shall be used as a basis of comparison for surface roughness determination."

Substitute: "The roughness comparison standards prescribed in ANSI B46.1 shall be used as a basis of comparison for surface roughness determination."

Preparing Activity:
Navy - OS
Project No. 1377-N797

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